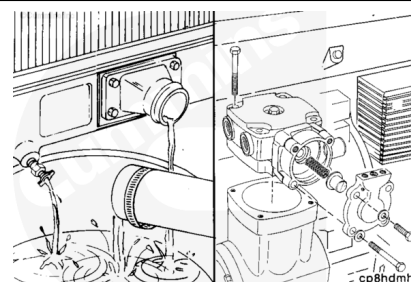


Trainings ▾

Disassemble

⚠ CAUTION ⚠

Use caution when draining coolant that coolant is not spilled or drained into the bilage area. Do not pump the coolant overboard. If coolant is not reused, it must be disposed of in accordance with local environmental regulations.

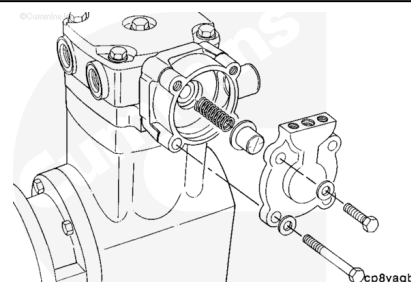


If the cylinder head is removed while the air compressor is on the engine, drain the engine coolant. Refer to the OEM troubleshooting and repair manual.

Since the valve plate, head and unloader body are indexible, marking of these parts is recommended to make sure they are reassembled in the proper orientation.

⚠ WARNING ⚠

The unloader body is installed with spring tension. Use care when removing to prevent personal injury. Always wear protective eye wear.



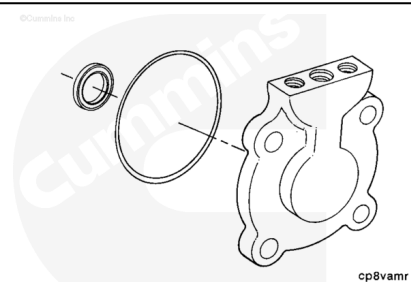
Hold the unloader valve body down and remove the four capscrews.

Remove the unloader valve body.

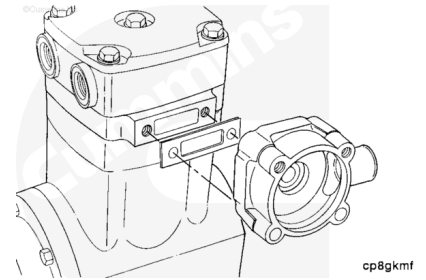
Remove the unloader valve spring.

Remove the unloader valve cap.

Remove the unloader body gasket and unloader valve cap rectangular ring seal.

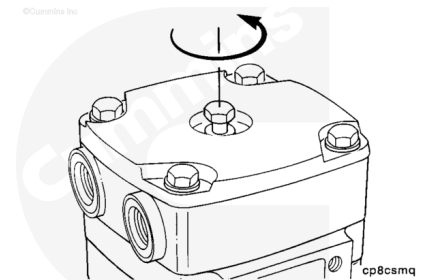


Remove the intake manifold and gasket.

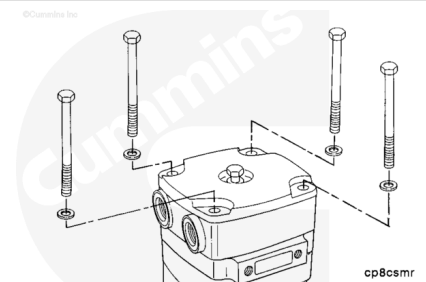


Loosen, but **do not** remove, the center head capscrew.

Mark the head for orientation during assembly.

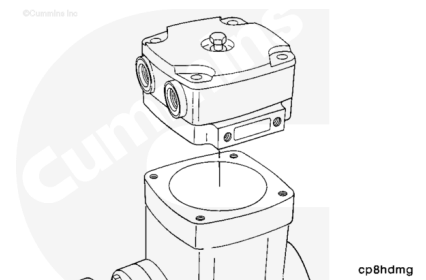


Loosen and remove the four corner head capscrews. Save the capscrews for reuse.



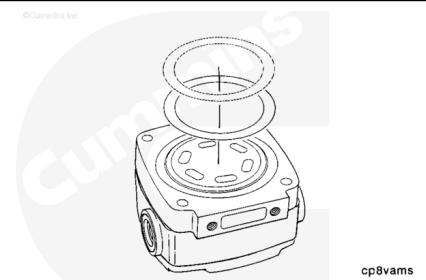
Remove the head, cover, and valve plate assembly and place it on a clean work surface with the intake valve facing upward.

If continuing with disassembly of the head, valve plate, and cover, be sure the work surface is clean. Grit pushed into the valve sealing surfaces by setting components on a dirty surface will cause a malfunction after reassembly.



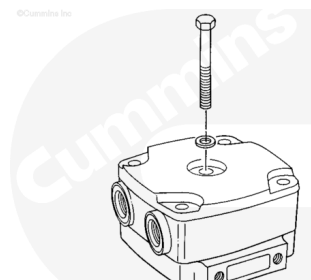
Some units have a press-fit intake valve retainer. If present, carefully remove it to prevent part damage.

Remove the intake valve.



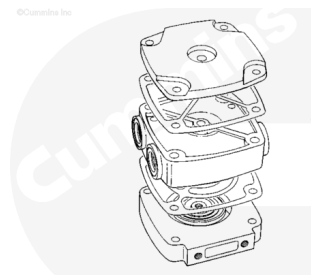
Turn the head assembly over and set it on a clean surface.
Remove the center capscrew. This capscrew can be reused.

Note : The center capscrew is shorter than the four corner capscrews.



cp8hdec

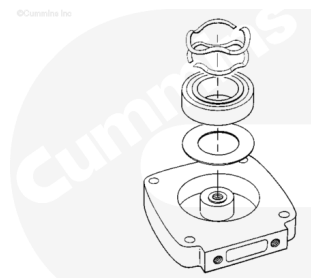
Remove the cover, cover gasket, head, and head gasket.



cp8cvmg

Remove the two wave washers, exhaust valve retainer and exhaust valve.

Note : The wave washers **must** be replaced.



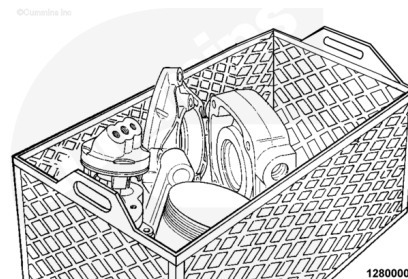
cp8wama

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing.

Soak the parts in a kerosene emulsion-based cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier can be contacted about solution concentration, temperature and soak time.

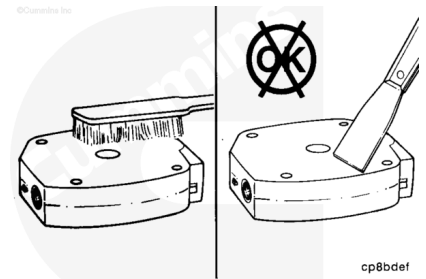


12800007

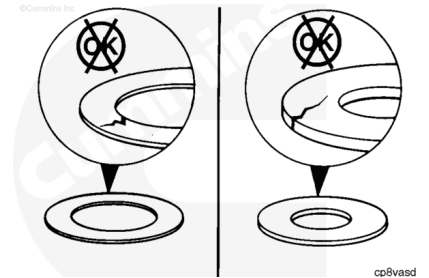
⚠ CAUTION ⚠

Do not use a scraper to remove carbon and scale, the sealing surfaces can be damaged.

The parts can be scrubbed with a stiff nonmetallic bristle brush.



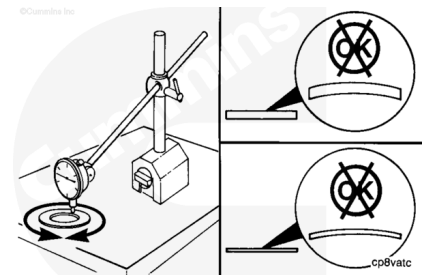
Inspect the intake and exhaust valves for cracks or damage.



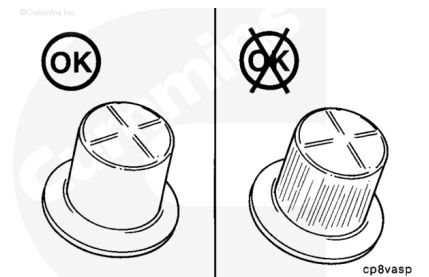
Measure the flatness of the intake and exhaust valves. Both valves **must** be flat within 0.03 mm [0.001 inch].

Replace valves if cracked, damaged, or **not** flat.

Note : Cummins® Turbo Technologies recommends new valves be installed.



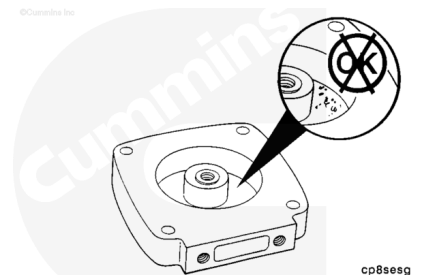
Inspect the upper part of the unloader valve cap where the rectangular V-seal operates. Check for scoring.



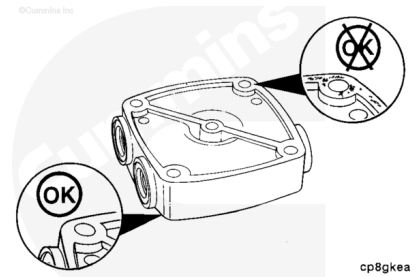
Inspect the valve seat surfaces.

Note : Inspection of the valve seats in the valve plate requires specialized equipment and is beyond the scope of field service.

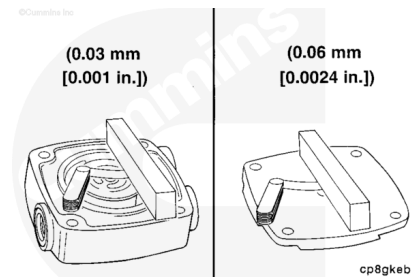
If the valve seat is damaged, or **cannot** be cleaned, a new valve plate is available in a service kit. Otherwise, a QE valve plate service assembly can be used.



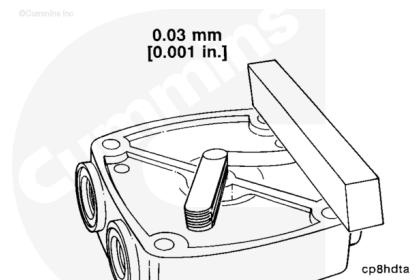
Gasket sealing surfaces **must** be clean and free of all old gasket material, carbon, rust, and other buildup. Surfaces **must** be free of scratches, gouges, burrs, and other deformities.



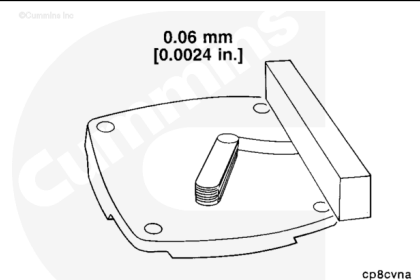
After making sure all gasket surfaces are clean and free of the above, inspect the head and cover for flatness. Use the flat plate and the feeler gauges.



Single cylinder heads **must** be flat within 0.03 mm [0.001 inch] between any two adjacent capscrew holes.



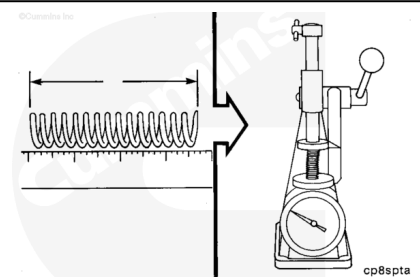
Single cylinder top cover **must** be flat within 0.06 mm [0.0024 inch] between any two adjacent capscrew holes and 0.10 mm [0.004 inch] total.



Use valve spring tester, Part Number 3375182, to check the unloader valve spring. Replace any spring that does **not** meet specifications.

Note : Cummins® Turbo Technologies recommends that new springs be installed.

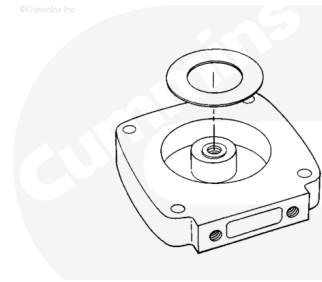
See the Compressor Spring Specifications chart shown at the front of this section.



Assemble

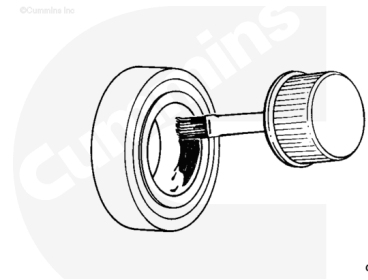
QE, Non-European

Install the exhaust valve over the post in the valve plate.



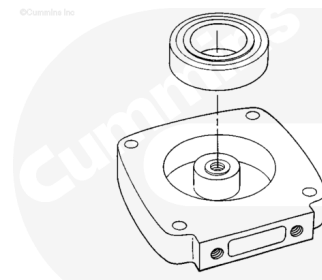
cp8vahk

Apply a thin coating of antiseize compound to the inside diameter of the exhaust valve retainer.



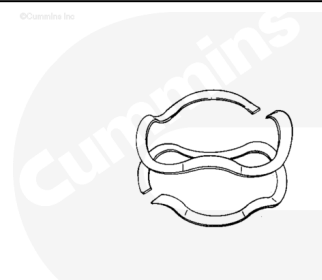
cp8rtwa

Slide the exhaust valve retainer over the valve plate. Make sure that the end of the retainer with the groove faces upward.



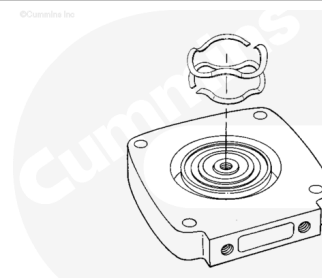
cp8rtha

Align the wave spring gaps 180 degrees from each other so they do **not** overlap.



cp8sphe

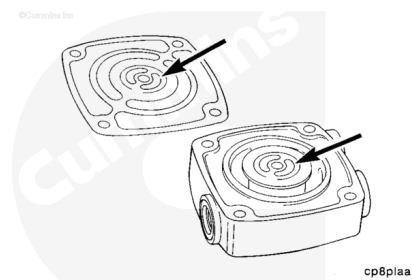
Place the wave springs in the retainer groove.



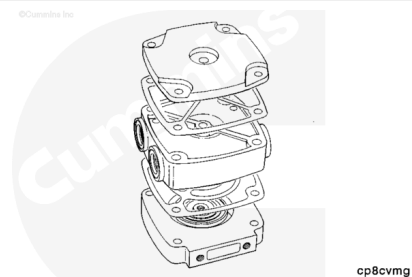
cp8sphf

Determine the final orientation of the valve plate (air intake location) and the head (coolant ports with respect to air inlet or manifold location). Align the kidney-shaped slots in the head with the kidney shaped slots in the gasket.

If orientation marks were made before disassembly, use them.



Assemble the cover, cover gasket, head, head gasket, and valve plate.

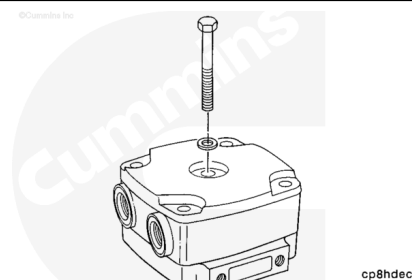


Make sure corner capscrew holes are aligned.

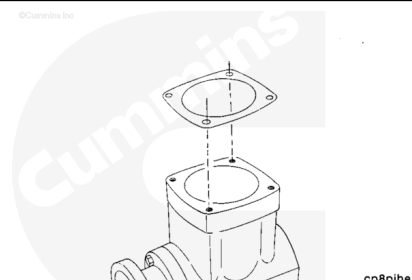
Lubricate the threads under the head.

Install the shorter capscrew with washer through the center hole.

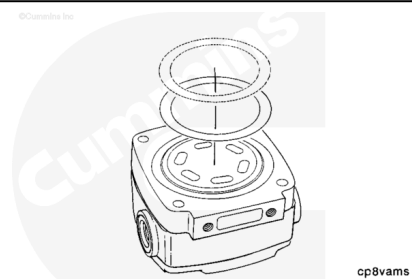
Torque Value: 14 n•m [124 in-lb]



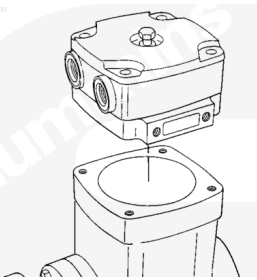
Install the valve plate gasket.



Carefully place the intake valve in the valve plate. Install the intake valve retainer.



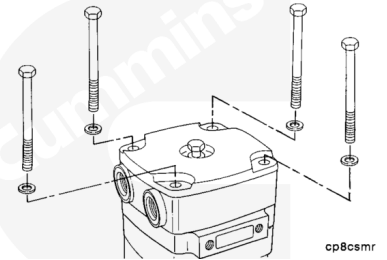
Install the valve plate assembly.



cp8hdmg

Lubricate the threads under the head and washer of the capscrews if initially installed.

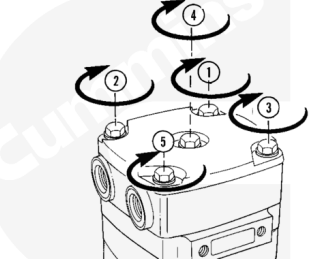
Install the four head capscrews and washers.



cp8csmr

Tighten all five capscrews.

Torque Value: 28 n·m [21 ft-lb]

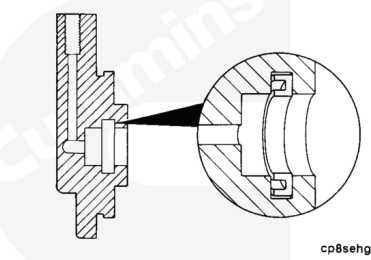


cp8cswb

Install the new rectangular V-seal into the unloader body.

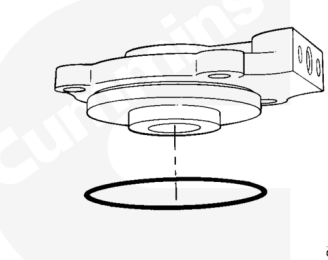
Note : The seal must be installed with the grooved side up.

Liberal lubricate the unloader valve bore above and below the rectangular ring seal with high temperature grease (Accrolube Lubrication Teflon™ grease or equivalent).



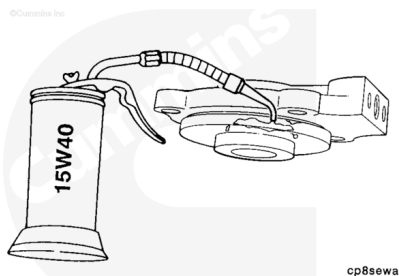
cp8sehg

Install a new o-ring seal on the unloader valve body.



cp8sehh

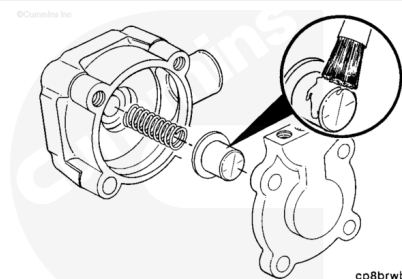
Use clean 15W-40 oil or Accrolube Lubrication Teflon Grease (or equivalent) to lubricate the seal.



Liberalily lubricate the unloader valve body bore and unloader cap with high temperature grease (Accrolube Lubrication Teflon™ grease or the equivalent).

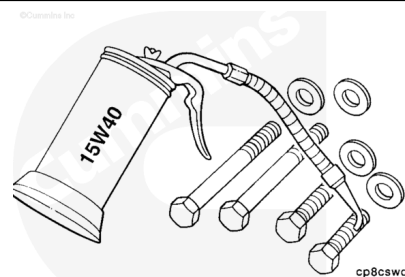
Install the unloader cap.

Install the unloader spring.



Lubricate the unloader screw threads and underhead with clean engine oil (SAE 15W-40), before installation.

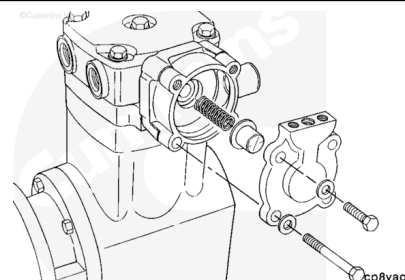
Note : The two unloader body screws must not be used to attach any brackets.



Assemble the unloader components and attach the unloader assembly to the valve plate with the four capscrews and washers.

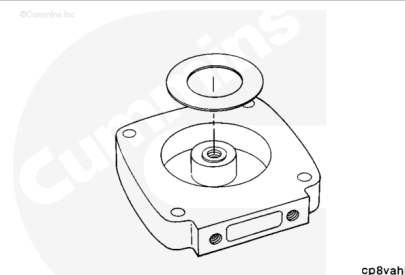
Note : The longer capscrews are used to mount the manifold to the air compressor.

Torque Value: 27 n•m [20 ft-lb]

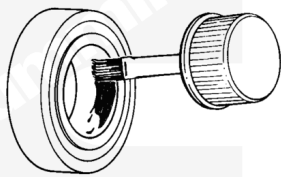


QE, European

Install the exhaust valve over the post in the valve plate.

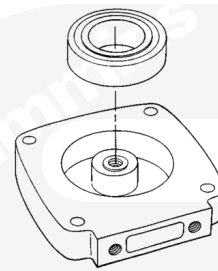


Apply a thin coating of antiseize compound to the inside diameter of the exhaust valve retainer.



cp8rtwa

Slide the exhaust valve retainer over the valve plate. Make sure that the end of the retainer with the groove faces upward.



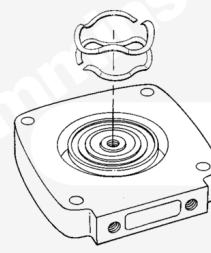
cp8rtha

Align the wave spring gaps 180 degrees from each other so they do **not** overlap.



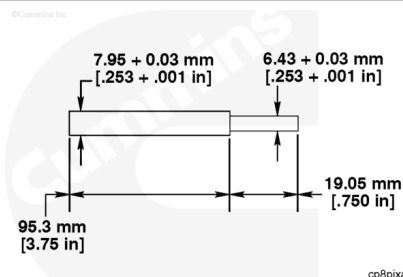
cp8sphe

Place the wave springs in the retainer groove.



cp8sphe

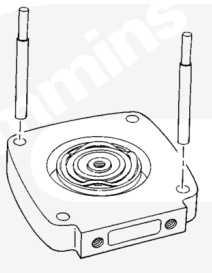
Fabricate or reuse the four guide pin tools.



cp8pixa

Insert the larger end of two guide pins in opposite corner holes of the valve plate.

©Cummins Inc.

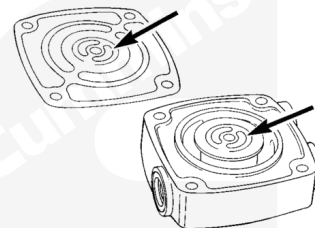


cp8pihc

Determine the final orientation of the valve plate (air intake location) and the head (coolant ports with respect to air inlet or manifold location). Align the kidney-shaped slots in the head with the kidney shaped slots in the gasket.

If orientation marks were made before disassembly, use them.

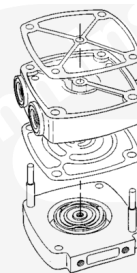
©Cummins Inc.



cp8piaa

Install the head gasket onto the guide pins (either side up, but with correct slot orientation). Install the head onto the guide pins with the kidney-shaped slots aligned and toward the valve plate. Install the cover gasket over the guide pins.

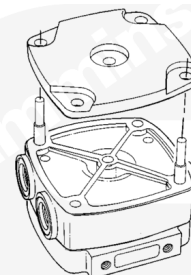
©Cummins Inc.



cp8gkhf

Assemble the cover.

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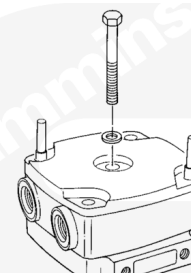


cp8cvhb

Install the shorter capscrew with washer, if initially installed, through the center hole.

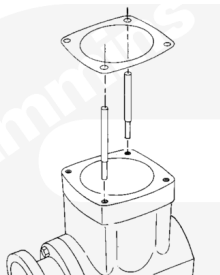
Torque Value: 14 n•m [124 in-lb]

©Cummins Inc.



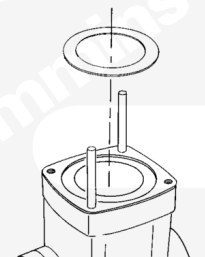
cp8cshh

Place the remaining two guide pins in the crankcase head capscrew holes (that will **not** interfere with the guide pins already in the head assembly). Install the valve plate gasket.



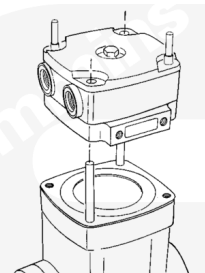
cp8pihd

Carefully place the intake valve on the crankcase, located by the valve plate gasket. Do **not** allow the valve to overlap the gasket.



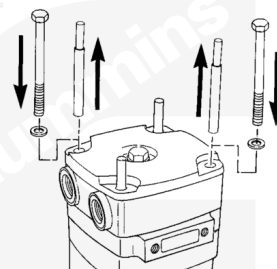
cp8vah1

Install the head assembly over the guide pins. Be careful **not** to disturb the location of the intake valve. The compressor will **not** work if the valve overlaps the gasket and is pinched.



cp8ndhb

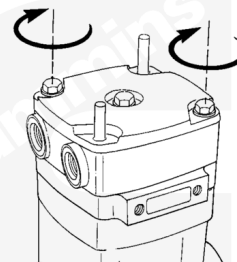
Carefully remove two of the guide pins and replace with two head capscrews and washers, if initially installed.



cp8pimb

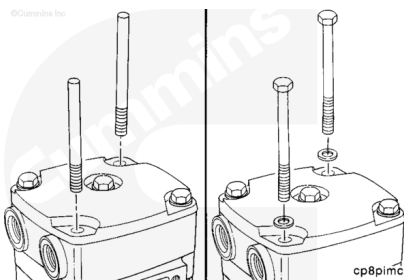
Tighten the two head capscrews.

Torque Value: 14 n•m [124 in-lb]



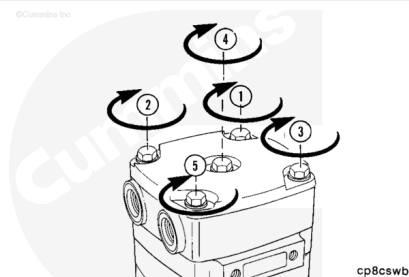
cp8cswa

Remove the remaining two pins and replace with two head capscrews and washers if initially installed.



Tighten all five capscrews.

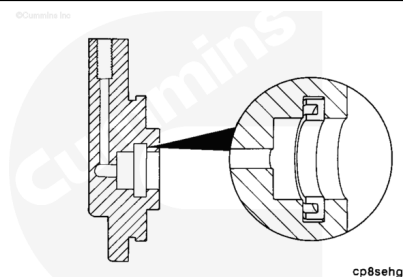
Torque Value: 28 n•m [21 ft-lb]



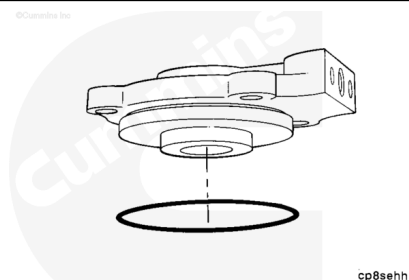
Install the new rectangular V-seal into the unloader body.

Note : The seal must be installed with the grooved side up.

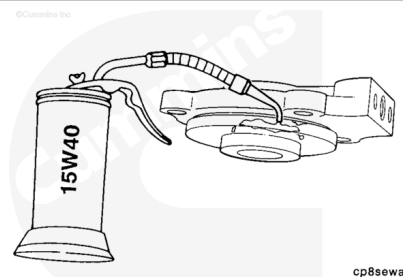
Liberally lubricate the unloader valve bore above and below the rectangular ring seal with high temperature grease (Accrolube Lubrication Teflon™ grease or equivalent).



Install a new o-ring seal on the unloader valve body.



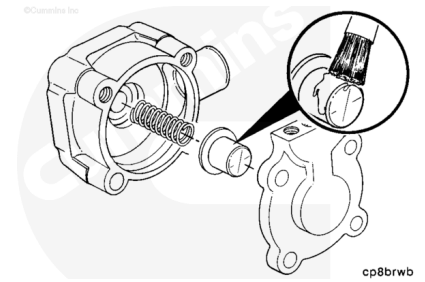
Use clean 15W-40 oil or Accrolube Lubrication Teflon™ grease (or equivalent) to lubricate the seal.



Liberally lubricate the unloader valve body bore and unloader cap with high temperature grease (Accrolube Lubrication Teflon™ grease or the equivalent).

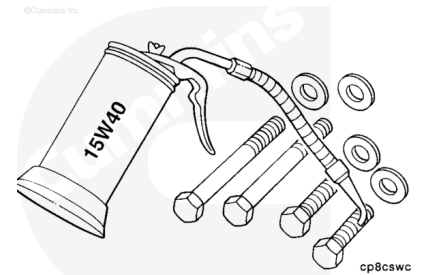
Install the unloader cap.

Install the unloader spring.



Lubricate the unloader screw threads and under the head with clean engine oil (SAE 15W-40) before installation.

Note : The two unloader body screws must not be used to attach any brackets.



Assemble the unloader components and attach the unloader assembly to the valve plate with the four capscrews and washers.

Note : The longer capscrews are used to mount the manifold to the air compressor.

Torque Value: 27 n•m [20 ft-lb]

